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Varendra University

Dept. of Computer Science & Engineering (CSE)

Addressing Mode and Mathematical Calculation

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

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Outline

1. Addressing Mode.
2. Memory Segmentation.
3. Segment and Offset.
4. Physical Address and Logical Address Calculation.
5. Address Line and Data Line Calculation.

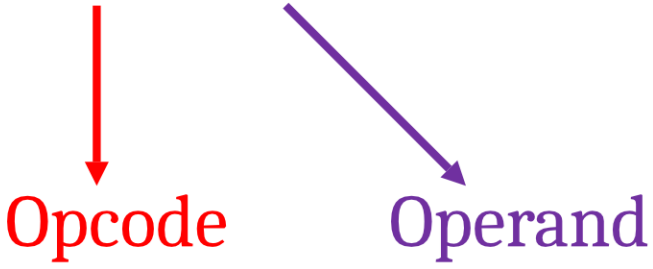
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Addressing Mode

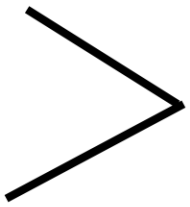
The manner in which operand is given an instruction is called addressing mode.

Example:

MOV AX, BX



MOV
ADD
SUB



Opcode (Operation)

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Addressing Mode

The manner in which operand is given an instruction is called addressing mode.

i) **Immediate mode:** Data is given in instruction.

```
MOV AL, 25H
MOV BX, 2000H
```

ii) **Register mode:** Data is given in register.

```
MOV AL, BL
MOV AX, BX
```

iii) **Direct mode:** Address is given in instruction.

```
MOV AX, [2000H]
```

iv) **Indirect mode:** Address is given in register.

```
MOV AL, [BX]
```

Addressing Mode

Indirect mode:

Register mode:

Physical address = Segment address + Effective address

MOV AL, [BX]

Base indexed:

Effective address = Base register + Index register

Base register = BX, BP

Index register = SI, DI

MOV AX, [BX+SI]

MOV AL, [BP+SI]

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Addressing Mode

Indirect mode:

Indexed mode:

Effective address = Index register + Displacement

MOV AX, [SI+2000]

Base mode:

Effective address = Base register + Displacement

MOV AL, [BP+2000]

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Addressing Mode

Indirect mode:

Base indexed displacement:

Effective address = Index register + Base register + Displacement

MOV AL, [SI+ BP+ 2000]

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Memory Segmentation

Segmentation is the process in which the main memory of the computer is logically divided into different segments.

Total memory size is divided into segments of various sizes is called **memory segmentation**.

➢ It helps in organizing and managing memory efficiently.

Advantage

- ❖ Efficient Memory Utilization.
- ❖ Protection.
- ❖ Faster Execution.

Memory Segmentation

- In memory, data is stored as bytes.
- 8086 microprocessor has 20 lines address bus.
  - Data bus 16-bit.
  - Address bus 20-bit.
- 20 address lines the memory can be addressed as,

$$2^{20}\text{bytes} = 1\,\text{MB}$$

- Address raging from 00000H to FFFFFH.
- Each register stores the base address (Starting Address) because segment resister cannot store 20 bits.

Segment and Offset

Memory address is specified using two components,

- **Segment:** Identifies the starting location (base address).
- **Offset:** Specifies the position within that segment.

8086 has 4 different types of segment,

1. Code Segment (CS).
2. Data Segment (DS).
3. Stack Segment (SS).
4. Extra Segment (ES).

} 16-bit

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Segment and Offset

Segment	Offset
CS	IP
DS	BX, DI, SI
SS	SP, BP
ES	DI

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Physical Address and Logical Address Calculation

Formula:

Physical Address = Base Address of Segment × 10H + Offset

Or

Physical address = Segment address of (Register Name) × 10H + Offset Address

Logical Address = Segment : Offset

Base Address (Starting Address)

Physical Address and Logical Address Calculation

Problem 1:

The content of following registers are,

CS = 1111H	DS = 1678H	SS = 6789H	ES = 1298H
IP = 6721H	BX = 7865H	DI = 1235H	BP = 7821H

Calculate the corresponding physical and logical address from address bytes in CS, DS, SS & ES.

Solution:

Segment: CS, DS, SS, ES

Offset: IP, BP, DI, BX

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Physical Address and Logical Address Calculation

Problem 1:

Solution:

Physical address of CS = Segment address of CS × 10H + Offset Address

= 1111H × 10H + 6721H

= 17831 H

Logical address of CS = Segment : Offset

= CS : IP

= 1111H : 6721H

➢ **Logical address is generated by CPU.**

Physical Address and Logical Address Calculation

Problem 1:

Solution:

Physical address of ES = Segment address of ES × 10H + Offset Address

= 1298H × 10H + 1235H

= 13BB5H

1 2 9 8 0
1 2 3 5
1 3 B B 5

Logical address of ES = Segment : Offset

= ES : DI

= 1298H : 1235H

Physical Address and Logical Address Calculation

Problem 1:

Physical address and Logical address of DS and SS.



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Physical Address and Logical Address Calculation

Problem 2:

Given a physical address and offset address. Calculate the corresponding segment address of ES.

Physical address = 13BB5H Offset address = 1235H

Solution:

Physical address = Segment address of ES × 10H + Offset Address

⇒ 13BB5H = Segment address of ES × 10H + 1235H

⇒ Segment address of ES = 1298H

Physical Address and Logical Address Calculation

Problem 3:

Calculate the physical address for following register.

SS = 3864H SP = 1735H BP = 4826H

Solution:

Case 1:

Physical address = Segment address of SS × 10H + Offset Address (SP)

= 3864H × 10H + 1735H

= 39D75H

Physical Address and Logical Address Calculation

Problem 3:

Calculate the physical address for following register.

SS = 3864H SP = 1735H BP = 4826H

Solution:

Case 2:

Physical address = Segment address of SS × 10H + Offset Address (BP)

= 3864H × 10H + 4826H

= 3CE66H

Physical Address and Logical Address Calculation

Problem 4:

Calculate the physical address for following instruction.

DS = 1000H SS = 7000H ES = 4000H

BP = 0100H SI = 0020H DI = 0200H BX = 0700H

Value (Displacement = 0500H)

Solution:

i) MOV AX, [BX][SI]

Physical address = Segment address of DS × 10H + [BX+SI]

= 1000H × 10H + [0700H+0020H]

= 10720H

Offset BX, SI = DS

Physical Address and Logical Address Calculation

Problem 4:

Calculate the physical address for following instruction.

DS = 1000H SS = 7000H ES = 4000H

BP = 0100H SI = 0020H DI = 0200H BX = 0700H

Value (Displacement = 0500H)

Solution:

ii) ADD AL, (BP+40H)

Physical address = Segment address of SS × 10H + (BP+40H)

= 7000H × 10H + (0100H+40H)

= 70140H

Offset BP = SS

Physical Address and Logical Address Calculation

Problem 4:

Calculate the physical address for following instruction.

DS = 1000H SS = 7000H ES = 4000H

BP = 0100H SI = 0020H DI = 0200H BX = 0700H

Value (Displacement = 0500H)

Solution:

iii) MOV CX, Values[BP]

Physical address = Segment address of SS × 10H +
[BP + Values]

= 7000H × 10H + (0100H+0500H)

= 70600H

Offset BP = SS

Physical Address and Logical Address Calculation

Problem 4:

Calculate the physical address for following instruction.

DS = 1000H SS = 7000H ES = 4000H

BP = 0100H SI = 0020H DI = 0200H BX = 0700H

Value (Displacement = 0500H)

Solution:

iv) MOV ES:[1000H], AL

Physical address = Segment address of ES × 10H + [1000H]

= 4000H × 10H + [1000H]

= 41000H

Physical Address and Logical Address Calculation

Problem 5:

Logical address is 8000H : 0000H. Calculate the physical address.

Solution:

Physical address = Segment address × 10H + Offset

= 8000H × 10H + 0000H

= 80000H

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Physical Address and Logical Address Calculation

Problem 6:

What is the physical address of 6th segment for location 250?

Solution:

Segment = 5000H **(Segment starts from 0. So, 5)**

Offset = 249 **(Start from 0. So, 249)**

= 00F9H **(Hexadecimal value)**

Physical address = Segment address × 10H + Offset

= 5000H × 10H + 00F9H

= 500F9H

Address Line and Data Line Calculation

Should be remembered,

$$K = 2^{10} \text{ (KB)}$$

$$M = 2^{20} \text{ (MB)}$$

$$G = 2^{30} \text{ (GB)}$$

$$T = 2^{40} \text{ (TB)}$$

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Address Line and Data Line Calculation

The memory units that follows are specified by the number of **words** times, the number of **bits** per word. How many **address lines** and input output **data lines** are needed in each case?

a) 8K × 16 b) 2G × 8 c) 16M × 32

d) 256K × 64 e) 64K × 8 f) 2T × 8

Solution:

8K × 16

Word Bit

- Data line is obtained from **"bit value"**
- Address line is obtained from **"Word value"**

Address Line and Data Line Calculation

The memory units that follows are specified by the number of **words** times, the number of **bits** per word. How many **address lines** and input output **data lines** are needed in each case?

a) 8K × 16 b) 2G × 8 c) 16M × 32

d) 256K × 64 e) 64K × 8 f) 2T × 8

Solution:

Let, address line is denoted by "A" and data line is denoted by "D".

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Address Line and Data Line Calculation

Solution:

a) 8K × 16

D = 16 (Data Line)

8K × 16

$= 2^3 \times 2^{10} \times 16$

$= 2^{13} \times 16$

A = 13 (Address Line)

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Address Line and Data Line Calculation

Solution:

b) 2G × 8

D = 8 (Data Line)

2G × 8

$= 2^1 \times 2^{30} \times 8$

$= 2^{31} \times 8$

A = 31 (Address Line)

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Address Line and Data Line Calculation

Solution:

c) 16M × 32

D = 32 (Data Line)

16M × 32

$= 24 \times 220 \times 32$

$= 224 \times 32$

A = 24 (Address Line)

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Address Line and Data Line Calculation

Solution:

d) 256K × 64

D = 64 (Data Line)

256K × 64

$= 2^8 \times 2^{10} \times 64$

$= 2^{18} \times 64$

A = 18 (Address Line)

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Address Line and Data Line Calculation

Solution:

e) 64K × 8

D = 8 (Data Line)

64K × 8

$= 2^6 \times 2^{10} \times 8$

$= 2^{16} \times 8$

A = 16 (Address Line)

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Address Line and Data Line Calculation

Solution:

f) 2T × 8

D = 8 (Data Line)

2T × 8

= 21× 240 × 8

= 241 × 8

A = 41 (Address Line)

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